

Day 302 – Environment Setup

Install TensorFlow, Keras, Create Your First Neural Network, and Manage Virtual Environments

Goal

Set up a professional Deep Learning environment and understand:

- What TensorFlow is
- What Keras is
- Why we use them
- How to install them
- How to create and use virtual environments
- How to build your first neural network

This day focuses on understanding the tools before building advanced Deep Learning projects.

Learning Outcomes

After completing Day 302, you will learn:

- ✓ What TensorFlow is
 - ✓ What Keras is
 - ✓ What a Virtual Environment is
 - ✓ Why Virtual Environments are important
 - ✓ How to create a Virtual Environment
 - ✓ How to activate and deactivate a Virtual Environment
 - ✓ How to install Deep Learning libraries
 - ✓ How to verify installations
 - ✓ How to create your first neural network
-

What is a Virtual Environment?

Definition

A Virtual Environment (venv) is an isolated Python environment that contains its own:

- Python packages
- Libraries
- Dependencies

without affecting the global Python installation.

Why Do We Use Virtual Environments?

Suppose:

Project A requires:

TensorFlow 2.x

Project B requires:

TensorFlow 3.x

If both are installed globally:

Version conflicts occur

Virtual environments solve this problem.

Real-World Example

Computer

|

|— Project A

| └─ venv

|

|— Project B

| └─ venv

|

└─ Project C

 └─ venv

Each project has its own dependencies.

How to Create a Virtual Environment

Open CMD in your project folder.

Example:

```
cd D:\Daily-GitHub
```

Create virtual environment:

```
py -3.11 -m venv dl_venv
```

Understanding the Command

`py -3.11`

Uses Python 3.11.

`-m`

Runs a module as a script.

`venv`

Python's built-in virtual environment module.

`dl_venv`

Name of the virtual environment folder.

Project Structure After Creation

Daily-GitHub/

|

| — dl_venv/

| | — Scripts/

| | — Include/

| | — Lib/

| | — pyvenv.cfg

|

| — project_files

How to Activate Virtual Environment

Windows CMD

```
dl_venv\Scripts\activate
```

Output:

```
(dl_venv) D:\Daily-GitHub>
```

The environment is now active.

Windows PowerShell

```
.\dl_venv\Scripts\Activate.ps1
```

Git Bash

```
source dl_venv/Scripts/activate
```

How to Check if Virtual Environment is Active

Run:

```
where python
```

Expected:

```
D:\Daily-GitHub\dl_venv\Scripts\python.exe
```

How to Install Packages Inside Virtual Environment

Once activated:

```
pip install tensorflow
```

or

```
pip install tensorflow numpy pandas matplotlib scikit-learn
```

The packages are installed only inside:

```
dl_venv
```

and not globally.

How to View Installed Packages

```
pip list
```

How to Save Installed Packages

```
pip freeze > requirements.txt
```

How to Install Packages from requirements.txt

```
pip install -r requirements.txt
```

How to Deactivate Virtual Environment

When your work is complete:

```
deactivate
```

Output:

```
D:\Daily-GitHub>
```

Notice:

```
(dl_venv)
```

disappears.

The environment is now turned off.

Typical Daily Workflow

Start Work

Open CMD:

```
cd D:\Daily-GitHub
```

Activate:

```
dl_venv\Scripts\activate
```

Output:

```
(dl_venv) D:\Daily-GitHub>
```

Work on Project

Run:

```
python app.py
```

or

```
python train_model.py
```

Finish Work

deactivate

What is TensorFlow?

Definition

TensorFlow is an open-source Deep Learning framework developed by Google.

Official Website:

[TensorFlow Official Site](https://www.tensorflow.org/)

What Does TensorFlow Do?

TensorFlow allows us to:

- Build Neural Networks
 - Train Deep Learning Models
 - Process Images
 - Process Text
 - Perform Predictions
-

Why Do We Use TensorFlow?

Without TensorFlow:

Thousands of mathematical calculations
must be written manually

With TensorFlow:

```
model.fit(...)
```

TensorFlow handles everything automatically.

What is Keras?

Definition

Keras is a high-level API built into TensorFlow.

It provides a beginner-friendly way to create neural networks.

Why Use Keras?

Without Keras:

Complex TensorFlow code

With Keras:

```
model = tf.keras.Sequential(...)
```

Much simpler.

TensorFlow vs Keras

TensorFlow	Keras
Engine	Interface
Powerful	Easy
Low-level	High-level
Handles computations	Handles model creation

Required Modules

```
pip install tensorflow numpy pandas matplotlib scikit-learn
```

Module Purpose

TensorFlow

Used for:

- Neural Networks
 - Deep Learning
 - Training Models
-

NumPy

Used for:

- Arrays
- Mathematical operations

Example:

```
import numpy as np
```

Pandas

Used for:

- CSV files
- Data analysis

Example:

```
import pandas as pd
```

Matplotlib

Used for:

- Charts
- Graphs
- Visualizations

Example:

```
import matplotlib.pyplot as plt
```

Scikit-Learn

Used for:

- Data preprocessing
- Dataset splitting
- Evaluation metrics

Example:

```
from sklearn.model_selection import train_test_split
```

Verify TensorFlow Installation

Run:

```
python
```

Then:

```
import tensorflow as tf
```

```
print(
```



```
tf.__version__  
)
```

Expected:

2.x.x

No errors means TensorFlow is installed correctly.

First Neural Network Concept

Dataset:

1 → 2

2 → 4

3 → 6

4 → 8

5 → 10

6 → 12

Pattern:

Output = Input × 2

The neural network learns this relationship automatically.

Neural Network Workflow

Input Data

|



Neural Network

|



Prediction

|



Loss Function

|



Weight Update

|



Next Epoch

Common Errors

Error

No matching distribution found for tensorflow

Cause

Unsupported Python version.

Example:

Python 3.14

Fix

Install:

Python 3.11

Create new environment:

```
py -3.11 -m venv dl_venv
```

Error

ModuleNotFoundError:

No module named tensorflow

Fix:

```
pip install tensorflow
```

Day 302 Summary

Today you learned:

- What a Virtual Environment is
- Why Virtual Environments are important
- How to create a Virtual Environment
- How to activate a Virtual Environment
- How to deactivate a Virtual Environment
- What TensorFlow is
- What Keras is
- Why TensorFlow and Keras are used
- How to install Deep Learning libraries
- How to verify installation
- The basic workflow of a neural network

This environment will be used throughout Days 303–330 for all Deep Learning, Computer Vision, NLP, and Flask AI projects.